For each question, show all related work and explain the solution.

1. (5 pts) Determine whether $168 \equiv 24(mod 16)$

By definition, $a \equiv b \pmod{n}$ iff $n \mid (a-b)$.

Compute the difference of $(a-b)$: $168 - 24 = 144$.

Now, check the divisibility by $n=16$: $144 \div 16 = 9$, so $16 \mid 144$.

Therefore, $168 \equiv 24 \pmod{16}$.

2. (15 pts) The octal expansion of an integer is $(4205)_8$. Write the decimal expansion, binary expansion and hexadecimal expansion.

Decimal Expansion: $(4205)_8 = 4 \cdot 8^3 + 2 \cdot 8^2 + 0 \cdot 8^1 + 5 \cdot 8^0 = 2048 + 128 + 0 + 5 = \boxed{2181.}$

Binary Expansion:

$2181 = 2 \cdot 1090 + 1$
$1090 = 2 \cdot 545 + 0$
$545 = 2 \cdot 272 + 1$
$272 = 2 \cdot 136 + 0$
$136 = 2 \cdot 68 + 0$
$68 = 2 \cdot 34 + 0$
$34 = 2 \cdot 17 + 0$
$17 = 2 \cdot 8 + 1$
$8 = 2 \cdot 4 + 0$
$4 = 2 \cdot 2 + 0$
$2 = 2 \cdot 1 + 0$
$1 = 2 \cdot 0 + 1$

$\boxed{(4205)_8 = 2181 = (1000\ 1000\ 0101)_2}$

Hexadecimal Expansion:

$2181 = 16 \cdot 136 + 5$
$136 = 16 \cdot 8 + 8$
$8 = 16 \cdot 0 + 8$

$\boxed{(4205)_8 = 2181 = (885)_{16}.}$

3. (15 pts) Use the Euclidean algorithm to write $gcd(21, 55)$ as a linear combination of 21 and 55. Be sure to show all steps that lead to your solution.

Euclidean Algorithm:

$55 = 2 \cdot 21 + 13 \longrightarrow 13 = 55 - 2 \cdot 21$
$21 = 1 \cdot 13 + 8 \longrightarrow 8 = 21 - 13$
$13 = 1 \cdot 8 + 5 \longrightarrow 5 = 13 - 8$
$8 = 1 \cdot 5 + 3 \longrightarrow 3 = 8 - 5$
$5 = 1 \cdot 3 + 2 \longrightarrow 2 = 5 - 3$
$3 = 1 \cdot 2 + 1 \longrightarrow 1 = 3 - 2$
$2 = 2 \cdot 1 + 0$

Back substitute to get the linear combination:

1.) $1 = 3 - 2$
2.) $1 = 3 - (5 - 3) = 2 \cdot 3 - 5$
3.) $1 = 2 \cdot (8 - 5) - 5 = 2 \cdot 8 - 3 \cdot 5$
4.) $1 = 2 \cdot 8 - 3(13 - 8) = 5 \cdot 8 - 3 \cdot 13$
5.) $1 = 5(21 - 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13$
6.) $1 = 5 \cdot 21 - 8(55 - 2 \cdot 21) = (5 + 16) \cdot 21 - 8 \cdot 55$
$= 21 \cdot 21 - 8 \cdot 55$

$\boxed{gcd(21, 55) = 1 = 21 \cdot 21 - 8 \cdot 55}$

4. (15 pts) Solve the congruence by first finding the inverse. $89x = 2 \ (mod\ 232)$

Verify the inverse exists:

An inverse exists if $\gcd(89, 232) = 1$.

$232 = 2 \cdot 89 + 54 \longrightarrow 54 = 232 - 2 \cdot 89$

$89 = 1 \cdot 54 + 35 \longrightarrow 35 = 89 - 1 \cdot 54$

$54 = 1 \cdot 35 + 19 \longrightarrow 19 = 54 - 1 \cdot 35$

$35 = 1 \cdot 19 + 16 \longrightarrow 16 = 35 - 1 \cdot 19$

$19 = 1 \cdot 16 + 3 \longrightarrow 3 = 19 - 1 \cdot 16$

$16 = 5 \cdot 3 + 1 \longrightarrow 1 = 16 - 5 \cdot 3$

$3 = 3 \cdot 1 + 0$

So, $\gcd(89, 232) = 1$

and an inverse exists.

Back substitute to find inverse:

1.) $1 = 16 - 5(19 - 16) = 6 \cdot 16 - 5 \cdot 19$

2.) $1 = 6(35 - 19) - 5 \cdot 19 = 6 \cdot 35 - 11 \cdot 19$

3.) $1 = 6 \cdot 35 - 11(54 - 35) = 17 \cdot 35 - 11 \cdot 54$.

4.) $1 = 17(89 - 54) - 11 \cdot 54 = 17 \cdot 89 - 28 \cdot 54$

5.) $1 = 17 \cdot 89 - 28(232 - 2 \cdot 89) = (17 + 56) \cdot 89 - 28 \cdot 232$

$= 73 \cdot 89 - 28 \cdot 232$

So, $89^{-1} \equiv 73 \pmod{232}$.

Solve for x: $89x \equiv 2 \pmod{232}$

$x \equiv 2 \cdot 73 \pmod{232}$

$\boxed{x \equiv 146 \pmod{232}.}$

5. (15 pts) Prove using mathematical induction that $\sum_{i=1}^{n} \left(\frac{1}{i(i+1)}\right) = \frac{n}{n+1}$, when n is a non-negative integer. Don't forget to begin with, "Let P(n) be the statement…".

Let P(n) be the statement $\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$, for a non-negative integer n.

Base Case $n = 0$: For $n = 0$, the left side is the empty sum, which is 0. The right side is $\frac{0}{0+1} = 0$. Thus, P(0) holds true.

Inductive Hypothesis: Assume P(K) is true for some $K \geq 0$. That is, $\sum_{i=1}^{K} \frac{1}{i(i+1)} = \frac{K}{K+1}$.

Inductive Step: We must show $P(K+1)$: $\sum_{i=1}^{K+1} \frac{1}{i(i+1)} = \frac{K+1}{K+2}$

$$\sum_{i=1}^{K+1} \frac{1}{i(i+1)} = \left(\sum_{i=1}^{K} \frac{1}{i(i+1)}\right) + \frac{1}{(K+1)(K+2)} = \frac{K}{K+1} + \frac{1}{(K+1)(K+2)}$$ by the inductive hypothesis.

$$\frac{K}{K+1} + \frac{1}{(K+1)(K+2)} = \frac{K(K+2)+1}{(K+1)(K+2)} = \frac{K^2+2K+1}{(K+1)(K+2)} = \frac{(K+1)^2}{(K+1)(K+2)} = \frac{K+1}{K+2},$$ which is $P(K+1)$.

Conclusion:

$\therefore$ By mathematical induction, P(n) holds for all non-negative integers n, that is,

$$\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$$ for all $n \in \mathbb{Z}^{0}$. □

6. (15 pts) Prove that $3^n < n!$, whenever n is an integer with $n \geq 7$.

Claim:

Let $3^n < n!$ for every integer $n \geq 7$.

Proof by induction:

Base case $n=7$: $3^7 = 2187$ and $7! = 5040$.
Thus, $3^7 < 7!$

Inductive Hypothesis:
Assume $3^k < k!$ for some $k \geq 7$.

Inductive Step: Then, $3^{k+1} = 3 \cdot 3^k < 3 \cdot k!$

Now, compare $(3 \cdot k!)$ with $(k+1)!$: $3 \cdot k! \leq (k+1) \cdot k! = (k+1)!$, since $3 \leq k+1$ for $k \geq 2$. Therefore, $3^{k+1} < 3 \cdot k! \leq (k+1)!$, which gives $3^{k+1} < (k+1)!$

$\therefore$ By induction, $3^n < n!$ for all integers $n \geq 7$. $\square$

7. (10 pts) Suppose that $\{a_n\}$ is defined recursively by $a_{n+1} = (a_n)^2 - 1$ and that $a_0 = 5$. Find a_3 and a_4.

$a_0 = 5$

$a_1 = (a_0)^2 - 1 = 5^2 - 1 = 24$

$a_2 = (a_1)^2 - 1 = 24^2 - 1 = 576 - 1 = 575$

$a_3 = (a_2)^2 - 1 = 575^2 - 1 = 330{,}625 - 1 = \boxed{330{,}624}$

$a_4 = (a_3)^2 - 1 = 330624^2 - 1 = 109{,}312{,}229{,}376 - 1 = \boxed{109{,}312{,}229{,}375}$

8. (10 pts) Give a recursive algorithm for computing na using **addition**, where n is a positive integer and a is a real number.

Multiplying $n \times a$ means adding a to itself n times.

So, recursively we can write the formula as $na = a + (n-1) \cdot a$

Base case $n=1$: $a + (n-1) \cdot a = a + 0 \cdot a = a$

Recursive Algorithm:

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procedure multiply (n: positive integer, a: real number)
    if n = 1 then
        return a
    else
        return a + multiply (n-1, a).
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